Rovem — Software Requirements Specification

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Version: Draft 0.5, superseding Draft 0.4. Title: Rovem.

Revision history

Version	Changes
Draft 0.2	Initial requirements, missions and design diagrams.
Draft 0.3	Adds the game flow, Knowledge and progression, power-up and unlock questions, the run prediction, and fully defined speeding-up, top-speed, coasting and braking behaviour. Sets mission start and pass rules, mission limits, Knowledge values, power-ups and the saving method, redesigns Mission 2 around the two kinds of friction, updates both diagrams, and settles the open items. Existing requirement IDs are unchanged; new requirements are numbered from FR-26.
Draft 0.4	Adds optional player accounts with progress saved to a cloud database, a local copy, and guest play. The Knowledge balance is calculated rather than stored, and every unlock is checked so that Knowledge never goes below zero. Clarifies how a wrongly answered question repeats. Revises FR-21, FR-23, FR-24, FR-25, FR-35, NFR-7 and NFR-8, and adds FR-41 to FR-44. Updates both diagrams. The database is described in the companion Data Model document.

Version	Changes
Draft 0.5	In Mission 3 a tow line holds the set speed on asphalt. Corrects the heavy-build statement for the brake line. Widens FR-31 to acceleration and F = ma. Defines the run predictions for Missions 1 and 2. States the power-up pickup positions and the grip limit on Motor boost. Adds the Physics Model Specification, decides hosting, and replaces the use case diagram. Revises FR-30, FR-31 and FR-32.

1. Introduction

1.1 Purpose

This document defines the requirements for the first version (MVP) of Rovem, a browser-based 3D game that teaches Grade 9 Kinematics and Dynamics through building and driving a rover.

Terms used in this document: MVP is the minimum viable product, meaning the first playable version. FR is a functional requirement, something the game must do. NFR is a non-functional requirement, a quality the game must have, such as performance. TBD means to be decided. Knowledge is the in-game currency, counted in points and earned by clearing questions: one cleared required question earns 10 points. A guest is a student who plays without an account. An account is an optional sign-in with an email address and a password.

1.2 Scope

Students build a rover from a few components, predict how it will behave, drive it through missions, and see the physics explained. The MVP has one terrain, three missions, one rover model built from parts, scenario questions, and Knowledge to unlock parts and missions. Students can play as guests or save their progress to an account. Everything else is listed in Section 6 as future work.

The physics is defined in full in the companion Physics Model Specification. Proposed component and surface values are in the Component Values sheet. The database is described in the companion Data Model document.

1.3 Learning objectives (MVP)

By the end, a student should be able to:

- Relate distance, time, speed and acceleration.
- Explain how mass and driving force affect acceleration (F = ma).
- Explain how friction affects motion and stopping.
- Explain why a body moves at constant speed when the forces on it balance (Newton's first law).
- Combine these ideas to solve a practical problem.

These objectives were checked against the Grade 9 Kinematics and Dynamics chapters by a Grade 9 physics teacher.

1.4 Learning loop

Predict → Experiment → Observe → Explain → Retry

2. Overall Description

2.1 Users

Primary users are Grade 9 students. Secondary users are teachers and evaluators who review the concept.

2.2 Constraints

- The game runs in a desktop web browser and needs no installation.
- The game is built by a single developer, so the scope must stay small.
- The physics must follow textbook formulas, not the default behaviour of the game engine.
- The physics uses g = 9.8 m/s², as the textbooks do. Numerical questions use g = 10 m/s² so answers are whole numbers, as in classroom practice.
- The interface language is English.

2.3 Assumptions

- Students have already met Newton's laws and friction in class, and the game reinforces and applies them.

- The terrain is flat, gravity is constant, and there is no air resistance in the MVP.

3. Functional Requirements

3.1 Game flow

ID	Requirement
FR-26	Each attempt follows the same order: build the rover, answer the mission's required questions not yet cleared, predict the run, drive, then view the results and the explanation.
FR-5	The student can change the build and retry any mission as many times as they like.

3.2 Rover customization

ID	Requirement
FR-1	The student selects one part for each slot: chassis, wheels and motor. The MVP has one rover model.
FR-2	Before a run, the game shows how the chosen parts affect the rover's mass, weight, rolling friction, grip limit, driving force, top speed, maximum acceleration and maximum braking deceleration on each surface of the mission.
FR-3	The values shown under FR-2 are calculated by the same physics model that drives the simulation.
FR-4	Each mission sets the rover's starting speed. The student does not change it.

3.3 Controls

ID	Requirement
FR-6	The student can steer left and right, apply throttle and apply brake. There is no reverse. Throttle and brake levels rise steadily from zero to full while the key is held and fall steadily back to zero when it is released. The maximum brake force is fixed for each rover model. Surface friction is a separate, external effect and is not part of the brake. If throttle and brake are held together, the brake applies and the throttle is released.
FR-7	Controls use a small number of simple inputs so they can later be adapted to touch screens.
FR-27	Steering changes the rover's direction, not its speed. A stopped rover does not turn. The driving screen shows speed in m/s and the direction of travel as an arrow.

3.4 Physics model

ID	Requirement
FR-8	Acceleration is net force divided by mass (a = Fnet / m), where net force is the driving force minus rolling friction and any braking force. Braking and friction only slow the rover; they cannot push it backward. A stopped rover stays stopped unless the driving force is greater than rolling friction.
FR-9	Rolling friction depends on the surface, the chosen wheels and the rover's weight (F = μR). Grip, set by the surface and the wheels, limits the driving and braking force the tyres can pass to the ground. Asking for more makes the wheels slip, and the screen shows it.

ID	Requirement
FR-10	Speed and distance are calculated with the equations of motion over short, fixed time steps in which force is constant, so results match the textbook relations exactly.
FR-11	Every implemented variable has the same effect every time. No part changes performance purely to make the game more entertaining.
FR-28	Each motor has a top speed. At top speed the motor only balances friction, so net force is zero and the rover moves at constant speed (Newton's first law).
FR-29	The physics uses g = 9.8 m/s², a fixed time step that does not depend on the frame rate, and no random values.

At every moment the rover is in one driving state. The state decides which forces act and therefore the acceleration:

State	When	Acceleration
Stopped	Speed is zero and the driving force cannot overcome friction, or the brake is held at zero speed	Zero
Driving	Throttle applied, below top speed	(Driving force − friction) / m
At top speed	Throttle held at the motor's top speed	Zero: forces balance, speed is constant
Cruising	Mission 3 only, while the tow line holds the rover, before the first brake input after the brake line	Zero: the tow balances friction, so speed stays at the set speed
Coasting	No throttle and no brake while moving, including after the Mission 2 power cut	−(friction) / m, slowing down
Braking	Brake applied while moving	−(brake force + friction) / m, until stopped

Formulas, control timings, the time step and the automated tests are defined in the Physics Model Specification.

3.5 Missions

Each mission defines an objective, a physics concept, a meaningful player decision, an observable result and feedback. Each mission puts one decision in front of the student, so only a few variables matter at once (NFR-6).

Mission	Objective	Concept	Decision that matters	Passed when
1. Motion Trial	From rest, reach the target within the time limit by a longer route on fast ground or a shortcut across slow ground.	Distance, time, speed, acceleration	Route, and motor force against mass and top speed	Target reached within 9 s

Mission	Objective	Concept	Decision that matters	Passed when
2. Surface Test	From rest, launch across an ice strip within the time limit. At the end of the ice the power cuts, and the rover must coast far enough into the sand.	Friction: grip and rolling friction	Wheels: grip against rolling friction. Mass and motor do not change the result	Ice crossed within 8 s, and the rover coasts at least 17.5 m into the sand
3. Stopping Challenge	The rover is towed at a set speed on asphalt and must stop with all of it inside the target zone. Braking is allowed only after the brake line, and the first brake input there releases the tow.	Force, friction and speed combined	Mass, brakes and grip, and when to brake	Rover stopped fully inside the zone

Mission parameters. These values come from simulating all 27 part combinations. They are the starting points for playtesting.

Parameter	Value	Reason
Mission 1 routes	100 m asphalt; 85 m sand	The faster route depends on the build (17 of 27 builds are faster on the shortcut)
Mission 1 time limit	9 s	The starting build passes on either route if driven cleanly, with 0.5–0.85 s to spare. Heavy builds with a small motor fail, which shows F = ma
Mission 2 ice strip	25 m	Long enough for grip differences to show in the time

Parameter	Value	Reason
Mission 2 ice time limit	8 s	All-terrain wheels pass with 0.36 s to spare; smooth wheels have too little grip and fail
Mission 2 coasting target	17.5 m into the sand	All-terrain wheels pass with 1.3 m to spare; spiked wheels have too much rolling friction and fail
Mission 3 approach speed	15 m/s, held by a tow line	Stopping distance (about 16–24 m) is much longer than the zone, so the student must plan where to brake. It is above the Large motor's 14 m/s top speed, so the tow, not the motor, holds it
Mission 3 surface	Asphalt	The Mission 3 stopping distances are calculated on asphalt
Mission 3 zone length	6 m	With a 2 m rover, the stopping window is 4 m, about 0.27 s of timing at 15 m/s: hard to hit by luck, fair with a prediction
Mission 3 brake line	23 m before the zone end	The starting build needs 18.8 m to stop. 4 of the 9 heavy builds need more than 23 m; they pass with Brake boost or lighter parts
Distance markers	Every 5 m	Lets the student turn a calculated distance into a place on the track

ID	Requirement
FR-12	The MVP contains these three missions, played on one terrain that has several surface types.
FR-13	Collectible items may be added for engagement, but they never replace the physics objective.

ID	Requirement
FR-30	In Mission 3 a tow line pulls the rover at its set speed. The first brake input after the brake line releases the tow and starts braking; braking before the brake line has no effect. After the release, letting go of the brake lets the rover coast. The throttle stays locked.
FR-31	Mission 1's explanation uses distance, time, speed, acceleration and F = ma, and calls the slower tiles "slow ground". Friction and grip are first explained in Mission 2.
FR-39	In Mission 2 the throttle drops to zero at the power-cut line at the end of the ice and stays locked for the rest of the attempt.

3.6 Feedback and results

ID	Requirement
FR-14	After each run, the game shows the measured results: distance, time, average speed, top speed reached, average acceleration, and stopping distance and time where relevant.
FR-15	The game explains why the outcome occurred, using pre-written explanations filled in with the run's own values, and shows the formula used, rather than only reporting success or failure.
FR-16	The game shows the student's predictions next to the actual result.

3.7 Scenario questions

ID	Requirement
FR-17	Each mission has three required scenario-based multiple-choice questions. They are shown before a run, as predictions, and never while the rover is moving.
FR-18	Questions test reasoning (for example "will it stop sooner or later on ice?"), not definition recall.
FR-19	Each question has a pre-written correct answer and a pre-written explanation. Numerical questions state "take g = 10 m/s²" so answers are whole numbers.
FR-20	A correctly answered question is marked "cleared" and not shown again.
FR-21	A wrongly answered question shows its explanation, which cannot be closed for 2 seconds, and is marked "to review". It reappears later as its variant: the same concept with a different scenario or numbers. If the variant is also answered wrongly, the original and the variant alternate until the question is cleared.
FR-22	A "Concepts cleared" screen lists cleared questions with their explanations.
FR-23	The game records how many times each question has been answered, so it knows whether the question was answered correctly on the first attempt.

ID	Requirement
FR-32	Before every attempt, the student also predicts the outcome of their own run: in Mission 1, the time to reach the target (under 8 s, 8 s to 9 s, or over 9 s); in Mission 2, whether the rover crosses the ice within 8 s and whether it coasts at least 17.5 m into the sand; in Mission 3, whether it stops short of, inside or past the zone. This prediction earns no Knowledge. Stopping distance for the chosen build is shown in the results, so the student can use it on the next attempt.
FR-33	Optional power-up questions appear at pickups on the course. The simulation pauses while one is shown. A correct answer gives a power-up for the current attempt; a wrong answer has no penalty.
FR-34	Optional unlock questions are available on the build screen and earn Knowledge for customization. Two unlock questions become available for each mission concept once that mission's required questions have been answered.

3.8 Knowledge and progression

ID	Requirement
FR-35	Knowledge is earned the first time a question is cleared. Required, power-up and unlock questions all earn Knowledge points, at the values below. There is no bonus for a first-attempt answer, so a wrong first answer costs nothing but time. The Knowledge balance is the points earned from cleared questions minus the cost of everything unlocked. No balance number is saved. The game saves only which questions are cleared and what is unlocked, and works out the balance from those whenever it is needed, so the balance always matches the progress.

ID	Requirement
FR-36	Knowledge unlocks rover parts and the next mission. A mission also requires the previous mission to be completed.
FR-37	At the start, one part per slot is available (standard chassis, all-terrain wheels, medium motor), and these parts can complete every mission. Unlocked parts are trade-offs, not strict upgrades.
FR-38	A power-up temporarily changes one variable the physics already uses, for the current attempt. Its effect is shown on screen as the old and new values. The power-ups offered in each mission are listed below.
FR-40	One power-up is active at a time; taking another replaces it. The mission timer pauses while a power-up question is shown.

Knowledge values. Costs are given in points and, beside them, in cleared questions, which is how a student experiences them.

Earned by	Points each	Points available	Reason
Required question (9 in total)	10	90	Core learning, so the main income
Power-up question (6 in total)	5	30	Optional; the power-up itself is the main reward
Unlock question (6 in total)	5	30	Optional; a second way to earn, so no student is stuck

Unlocks	Cost in points	In cleared questions	Reason
Mission 2, Mission 3	20 each	2 required questions	Mission 1's three required questions earn 30 points before the first drive, so Mission 2 is already affordable. One wrong answer never blocks progress
Each locked part (6 in total)	15 each	1½ required questions, or 3 optional ones	Same price for every part, because parts are trade-offs, not upgrades

Unlocking everything costs 130 of the 150 points available: for example, all 9 required questions plus 8 of the 12 optional ones. A student who clears Mission 1's three required questions can open Mission 2 and still has 10 points left.

Power-ups. Each one changes a single variable, so the student sees one cause and one effect.

Mission	Power-up	Change	What the student sees
1	Motor boost	Driving force +20%	More force gives more acceleration (F = ma). On slow ground the wheels can slip, so part of the extra force is lost
1	Lighter load	Mass −10%	Less mass gives more acceleration from the same force
2	Grip tyres	Wheel grip +30%	Grip limits how hard the rover can accelerate on ice
2	Smooth bearings	Rolling friction −30%	Less rolling friction lets the rover coast further
3	Brake boost	Brake force +20%	More brake force stops a heavy rover sooner, unless grip is the limit
3	Grip tyres	Wheel grip +30%	When grip is the limit, more grip shortens the stop

Pickups sit where the change matters: Motor boost and Lighter load side by side near the start of each Mission 1 route, so the student chooses force or mass on either route; Grip tyres at the start and Smooth bearings at the end of the Mission 2 ice strip; and Brake boost and Grip tyres on two lanes before the Mission 3 brake line. With the right power-up, a build that would otherwise fail can pass: 5 of the 7 failing builds in Mission 1, all 18 in Mission 2, and all 4 in Mission 3, with Brake boost.

3.9 Progress saving and accounts

ID	Requirement
FR-24	The game saves, after each answer, the number of times the question has been answered and, when it is cleared, the time it was cleared. It saves each unlock after the unlock, and a mission's completion when the mission is cleared. A completed mission is a safe point: the game resumes from it in the next session. A question's status (cleared, to review or unanswered), whether it was right the first time, and the Knowledge balance are not saved. The game works them out from this data each time they are needed.
FR-25	A signed-in student's progress is saved to their account in a cloud database, with a copy in the browser's local storage. A guest's progress is kept in local storage only. Answers and mission completions made without a connection are sent to the account when the connection returns.
FR-41	A student can play as a guest, or sign up and sign in with an email address and a password. A signed-in student can sign out, reset their password, and delete their account together with all of its progress. A new account starts with Mission 1 and the three starting parts unlocked.
FR-42	A student can read only their own progress. A student can add or update only their own answers and mission completions. Unlocks are recorded only through the check in FR-44.

ID	Requirement
FR-43	At first sign-in, a guest's progress is copied into the account if the account has none. If the account already has progress, the student chooses which of the two to keep, so Knowledge is never counted twice. When two copies of the same account differ, for example after playing on two devices, the higher attempt count and the earlier clearing time are kept for each question.
FR-44	An unlock is recorded only if the Knowledge balance covers its cost and, for a mission, the previous mission is completed. The Knowledge balance therefore never goes below zero. A signed-in student needs a connection to unlock; a guest unlocks on the device under the same rule.

4. Non-Functional Requirements

ID	Requirement
NFR-1	Platform: runs in current desktop browsers. Mobile browser support is a later phase.
NFR-2	Performance: playable at 30 frames per second or more on a laptop with integrated graphics (proposed target). Physics results do not depend on frame rate.
NFR-3	Load time: low-poly models and compressed assets keep the first download small (proposed limit: 10 MB) so the game loads on slow connections.
NFR-4	Physics accuracy: automated tests compare simulation output with hand-calculated values, with a relative error below 10^{-9}. The test cases are listed in the Physics Model Specification.

ID	Requirement
NFR-5	Extensibility: components, surfaces, terrains, missions, questions and power-ups are stored as data, so new ones can be added without rebuilding the game.
NFR-6	Clarity: only a few adjustable variables are shown at once, and complexity increases gradually across missions.
NFR-7	Offline fallback: once loaded, the MVP works without an internet connection, because all explanations are pre-written. Signing in, saving to the account and unlocking while signed in need a connection; other progress is kept on the device and sent to the account later (FR-25, FR-44).
NFR-8	Privacy: the MVP collects only what accounts need, which is an email address and the student's progress. Guests provide no personal data. Any playtest analytics are anonymous and used only with the school's consent. Accounts for real students are used only with the school's consent.

5. Validation

- **Physics:** automated tests per NFR-4.
- **Balance:** all 27 part combinations are simulated against the mission limits (Component Values sheet). The starting build passes all three missions (FR-37), and every mission has builds that fail for a reason the explanation can name. Values are then tuned in playtesting.
- **Progress data:** automated tests check the Knowledge calculation (FR-35), the unlock rule (FR-44) and the merge rules (FR-43) against hand-worked cases.
- **Learning (optional):** a short pre/post question set (4–5 questions) and a small trial with a few students, recording first-attempt correctness per concept. First-attempt data alone is weak evidence, since students may learn from the run itself.

6. Future Work (out of MVP scope)

- Three default rovers, and rovers designed by students using mass, weight and speed.
- Brakes become a fourth part slot, so brake force comes from a part.
- Student-made levels and challenges: terrain, planet and objectives, each with a visible cost or effect, so students see how environments change motion.
- Modifiers: in-game events (for example cannons, aliens or space weather) that each change one variable, such as mass, force, friction or velocity. Each shows its effect on screen.
- Momentum and impulse, which are in the Grade 9 Dynamics chapter, introduced through modifiers that change velocity by a force acting over a short time.
- Weight vs. mass through different gravity, only if it stays simple for Grade 9.
- More components, terrains and missions.
- AI helper that rephrases explanations and answers follow-ups using the run's data. It needs a server-side proxy so the API key is not exposed in the browser, and pre-written explanations stay as fallback.
- Mobile browser version with touch controls.
- Sharing of student-made levels.

7. Open Items

Item	Status
Syllabus check	Done: checked by a Grade 9 physics teacher. Textbooks use g = 9.8 m/s²
Component and surface values, motor top speeds	Set in the Component Values sheet; confirmed in playtesting
Control timings and physics time step	Set in the Physics Model Specification: 1/120 s step, throttle build-up 1.0 s, brake build-up 0.5 s; confirmed in playtesting
Mission limits, zone and brake line	Set (3.5); confirmed in playtesting
Knowledge values, unlock costs and power-ups	Set (3.8); confirmed in playtesting

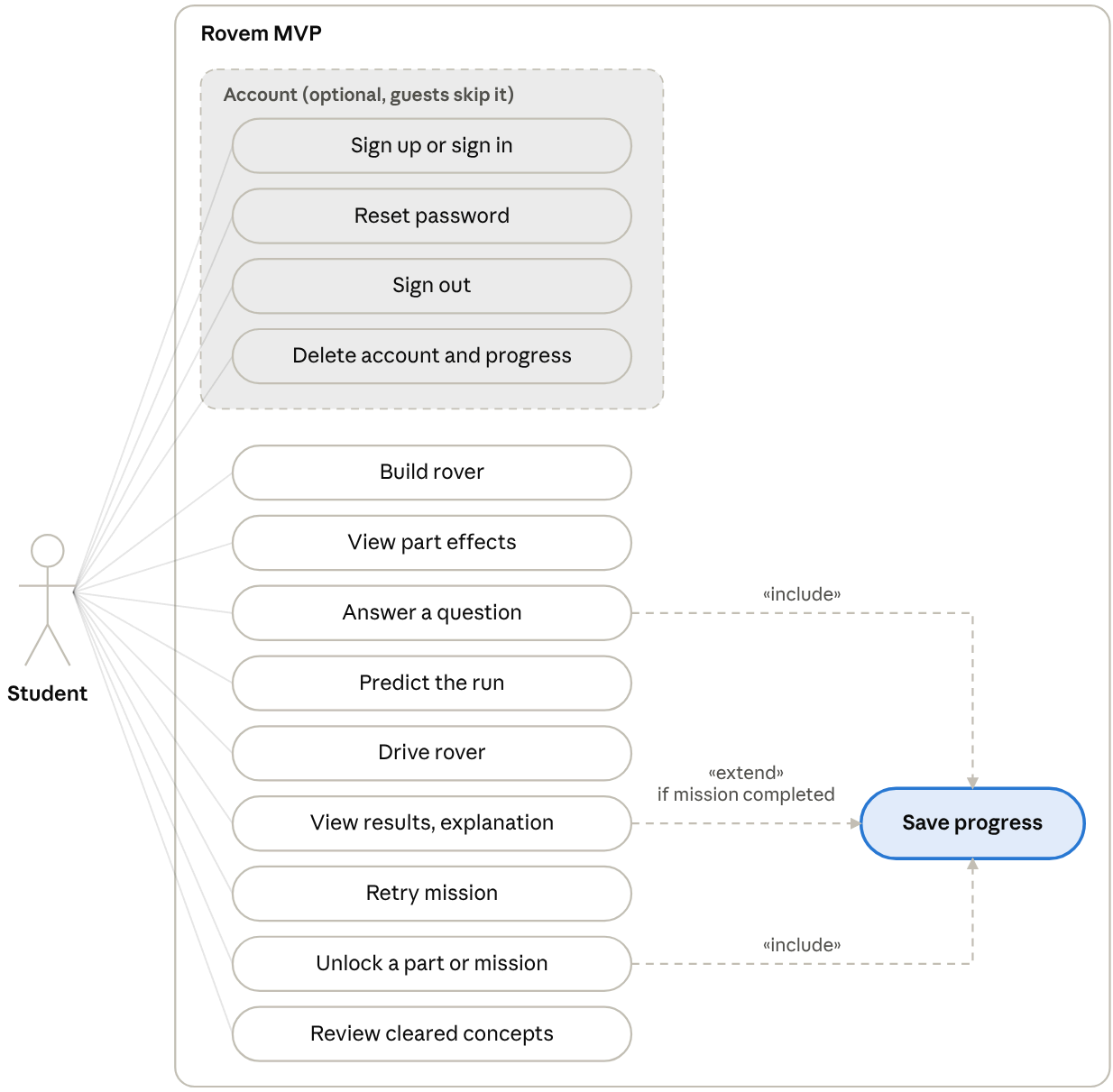
Item	Status
Question content: 9 required, 6 power-up and 6 unlock questions, each with one variant	To be written
Tech stack and hosting	Database: PostgreSQL on Supabase. Hosting: Nginx on a Linux server. The rest of the stack is provisional: see the Tech Stack document
Saving method	Decided: cloud account with a local copy and guest play (FR-24, FR-25, FR-41)
Account email handling and school consent	To be confirmed with the supervisor before any student pilot

8. Design Diagrams

UML (Unified Modeling Language) is a standard set of diagrams for showing how software is designed before it is built. The two diagrams below answer two questions: what can a student do in the game (use case diagram), and what things does the game contain and how are they connected (class diagram).

8.1 Use case diagram

A student has thirteen interactions with Rovem, and three of them save progress

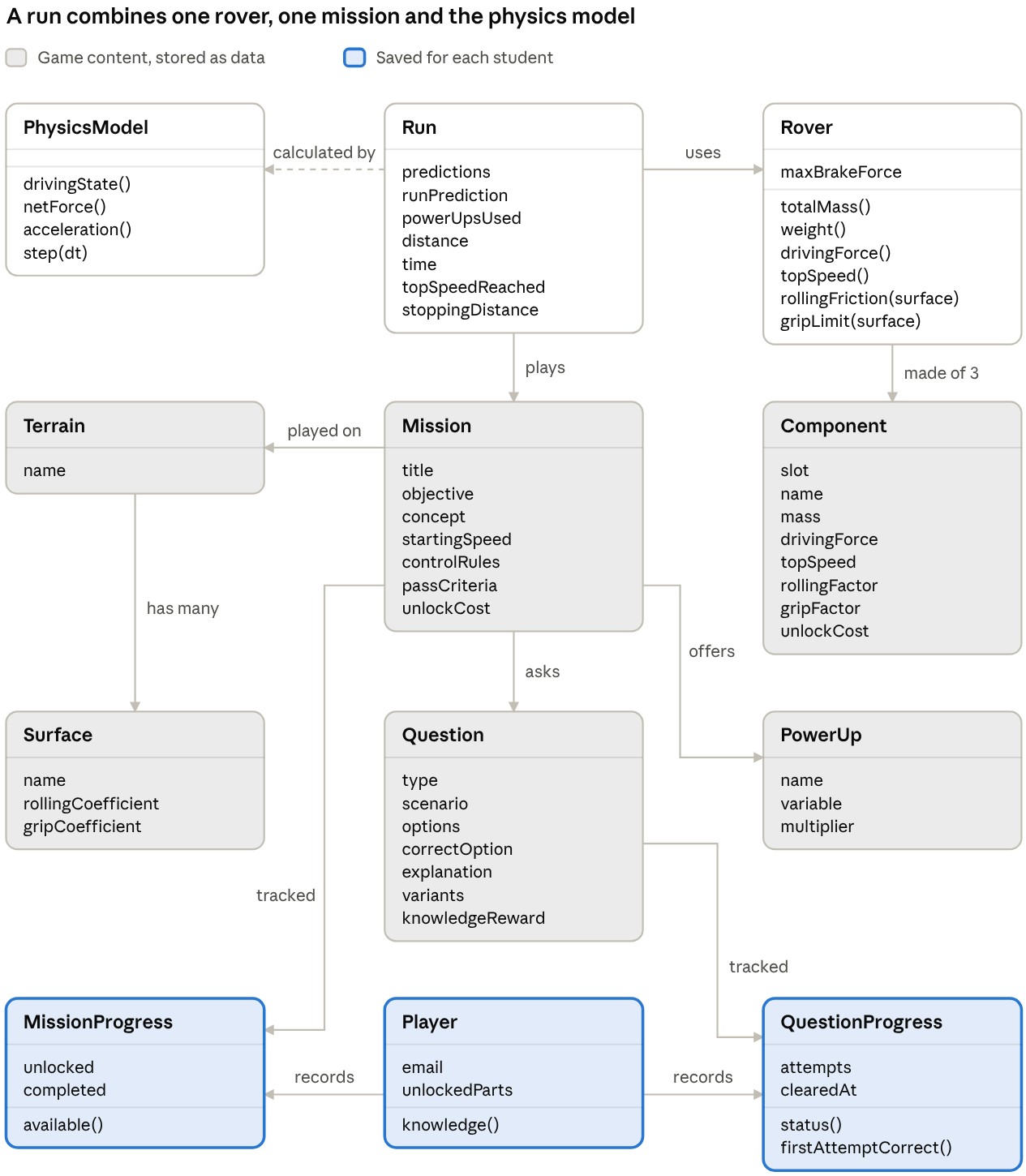


use case diagram · 1 actor, 14 use cases

The student has thirteen interactions with the game. The four account actions are optional: a guest plays without an account and never needs them (FR-41). Answering any question and unlocking a part or mission always save progress (FR-24). Viewing results saves progress only when the mission has just been completed, which makes it a safe point. A signed-in student's progress is saved to their account (FR-25); a guest's stays on the device.

How to read it: the stick figure is the student and the large box is the game. Each oval is one thing the student can do, and a line from the student to an oval means the student can do it. The shaded group holds the optional account actions. A dashed arrow marked «include» means the action it starts from always triggers the action it points to: answering a question always saves progress. A dashed arrow marked «extend» means the action it starts from is added to the action it points to only when its condition holds: progress is saved after viewing results only if the mission was completed.

8.2 Class diagram



class diagram · 12 classes

Shaded classes are stored as data (NFR-5). Classes outlined in blue are saved for each student (FR-24). The maximum brake force is a fixed attribute of the rover model (FR-6). Top speed comes from the motor (FR-28), and each surface has separate rolling-friction

and grip values (FR-9). The Knowledge balance, a question's status and whether it was right the first time are not saved. They are worked out from the saved data whenever they are needed (FR-35).

How to read it: each box is one kind of thing in the game. The name is at the top. Below it are the pieces of information it holds, or, when followed by brackets such as totalMass(), a calculation it can perform. An arrow joins two boxes, and its label tells you how to read it, for example "Run uses Rover" or "Mission asks Question". A dashed arrow means one box relies on the other to work something out: "Run is calculated by PhysicsModel".

Class	In plain words
Rover	The vehicle the student builds and drives.
Component	One part (chassis, wheels or motor) that changes the rover's mass, friction, grip, driving force or top speed. Some cost Knowledge to unlock.
Terrain	The ground the missions are played on.
Surface	One type of ground on the terrain, with its own rolling friction and grip.
Mission	One challenge, with its goal, physics concept, starting speed, control rules, pass rule and unlock cost.
MissionProgress	Whether a student has unlocked a mission and whether they have completed it. A completed mission is the safe point.
Run	One attempt at a mission: the student's predictions, any power-ups used, and the measured result.
PhysicsModel	The part that decides the driving state and calculates force, acceleration, speed and distance from the formulas.
PowerUp	A temporary change to one physics variable (for example driving force × 1.2), offered by a mission and earned with a correct answer.

Class	In plain words
Question	One scenario question (required, power-up or unlock) with its answer, explanation, variants and Knowledge reward.
QuestionProgress	How many times the student has answered a question and when they cleared it. Whether it is cleared, to review or unanswered, and whether it was right first time, are worked out from those two facts.
Player	The student's account: the sign-in email and the parts they have unlocked. There is no saved Knowledge total; the balance is worked out from their progress when needed.